



Infiniband & High-Speed Interconnects

Feeding Distributed AI at Microsecond Scale

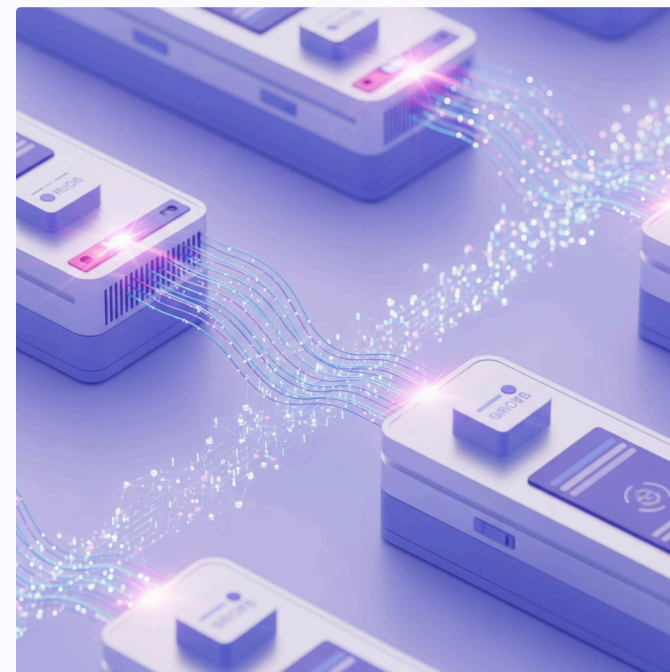
A technical deep dive for infrastructure engineers and AI system architects

Why Interconnects Matter for AI

The foundation of distributed training performance rests on network fabric quality:

- Collective operations (All-Reduce/All-Gather) dominate training time
- **p99 latency** and **jitter** drive scalability more than averages
- Poor fabrics → idle GPUs, unstable throughput, wasted investment

Our goal: maximize **bisection bandwidth** and predictability at scale



Infiniband vs Ethernet (RoCEv2)

Infiniband (IB)

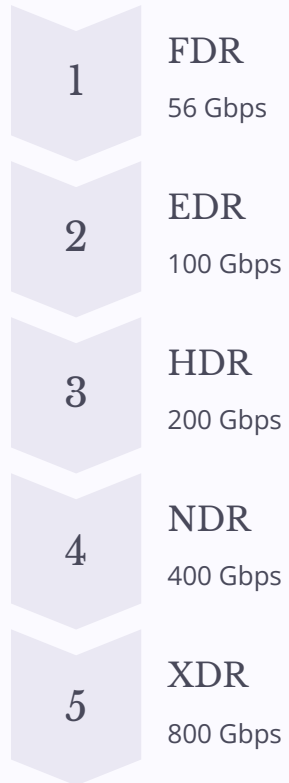
- Purpose-built for HPC workloads
- Hardware-based flow control (credit-based)
- Deterministic low-latency performance
- Mature collective operations support
- Specialized management tools

Ethernet + RoCEv2

- Ubiquitous enterprise technology
- Flexible deployment options
- Requires tight QoS/congestion tuning
- Broader ecosystem compatibility
- Typically lower hardware costs

Choice depends on scale, budget constraints, operations team skills, and multi-tenant infrastructure needs. Both can deliver state-of-the-art performance if engineered correctly.

IB Generations & Speeds



Engineering best practices:

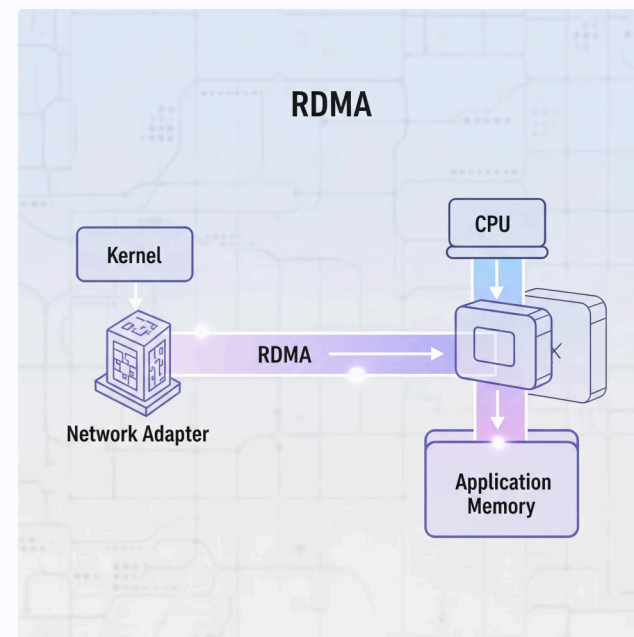
- Scale links **east-west**; plan upgrade paths early in your architecture
- Keep optics/cabling SKUs standardized for simpler operations
- Balance between current needs and future-proofing your infrastructure

RDMA Fundamentals

Remote Direct Memory Access

- Zero-copy data transfers between network-connected systems
- Kernel bypass eliminates OS overhead
- Minimal CPU utilization during transfers
- Verbs API model with send/receive, RDMA read/write operations
- Completion queues for asynchronous operation

NIC/adaptor offloads reduce host overhead and critical tail latency



Essential foundation for:

- NCCL communication library
- MPI in traditional HPC
- Parameter servers in distributed training

AI Accelerants on Infiniband



NCCL over IB

Optimized collective communication operations leveraging RDMA primitives with specialized algorithms for AI workloads



GPUDirect RDMA

Direct NIC ↔ GPU transfers bypass host memory, reducing latency and host memory bandwidth bottlenecks



In-Network Compute

SHARP-style offloads reduce traffic for reductions by performing aggregation in the network fabric

Placement matters: Align GPU ↔ NIC PCIe paths and NUMA affinity for optimal performance. Co-locate resources to minimize cross-node communication.

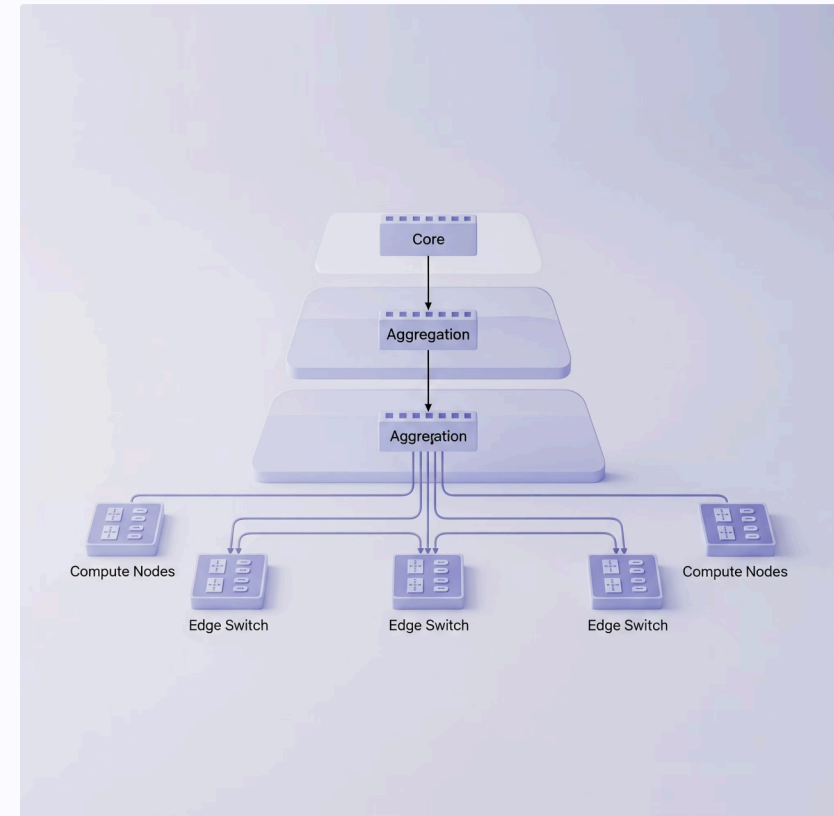
Fabric Topology & Routing

Network Topologies

- **Fat-Tree/Clos:** Non-blocking fabrics with full bisection bandwidth
- **Dragonfly:** Optimized for scale & cost with controlled oversubscription

Traffic Management

- Adaptive routing to dynamically dodge hot spots
- ECMP-like path diversity for traffic distribution
- QoS / Service Levels (SLs) to separate traffic classes



Always **measure & enforce oversubscription ratios** explicitly in your architecture. Document them as part of your baseline configuration.



Ethernet/RoCEv2 Essentials

Lossless Operation

Implement Priority Flow Control (PFC) to create lossless traffic classes, with congestion feedback via ECN/DCQCN mechanisms

End-to-End Consistency

Harmonize MTU, buffer allocations, and queue maps across the entire fabric to prevent bottlenecks and packet drops

Hardware Selection

Deploy deep buffers on spine/leaf switches to absorb bursty collective operation traffic patterns

Validation Testing

Continuously verify with both synthetic benchmarks (iperf3) and application-level tests (nccl-tests)

Hardware Building Blocks



HCA/NICs & DPUs

Modern RDMA adapters with offload capabilities; DPUs for isolation and virtualization support

Standardize firmware baselines and qualification matrices across your environment to reduce troubleshooting complexity and ensure consistent performance.



Switches

Leaf/Spine switches with large buffers, telemetry capabilities, and consistent firmware



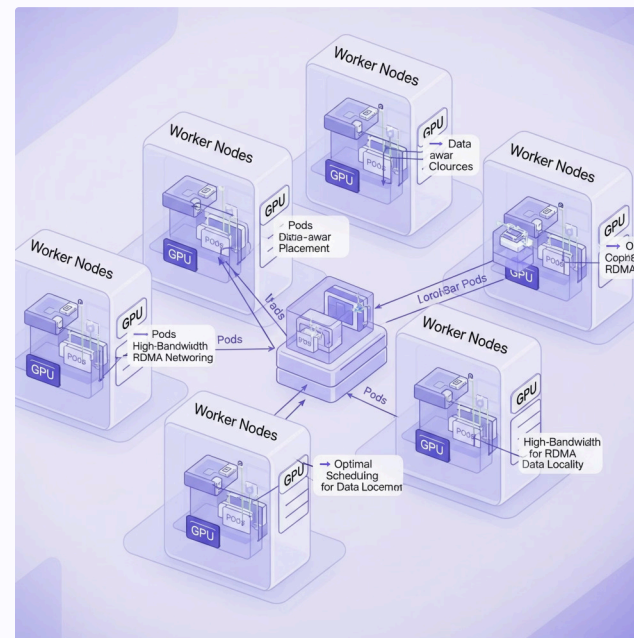
Cabling Infrastructure

DAC/AOC/optics selection based on reach requirements, power constraints, and cost considerations

Kubernetes Integration

Modern AI infrastructure requires container orchestration with high-performance networking:

- **RDMA device plugin** + **SR-IOV** for pod-level RDMA access
- Container Network Interface with eBPF datapaths for performance
- Pin pods to GPU/NIC-local nodes to minimize cross-node traffic
- Container images with MOFED/RDMA libraries aligned to host drivers
- Job schedulers (Kubernetes + Volcano/Ray) with network topology awareness



- ❏ Ensure your container image includes RDMA libraries that match the version on your host system to prevent compatibility issues.

Observability & Troubleshooting

Infiniband Diagnostics

Tools for fabric health monitoring:

- ibdiagnet for topology validation
- perfquery for performance statistics
- ibstat for port status
- Fabric managers and telemetry systems

Hardware Topology

Verify system configuration:

- nvidia-smi topo -m for GPU/NIC relationships
- PCIe topology validation
- NUMA affinity checks

RoCE Monitoring

Critical metrics to track:

- PFC pause frame counts
- ECN marking statistics
- Queue depth measurements
- Buffer utilization

Workload testing should include both **nccl-tests** (all_reduce_perf) for collective operations and low-level benchmarks like ib_write_bw and ib_send_lat for baseline performance verification.

Reliability & Operations

Redundancy Architecture

- Dual-spine topology with redundant physical paths
- Fast reroute capabilities on link or node failure
- Fabric health monitoring with proactive alerts

Change Management

- Strict change control processes
- Staged rollouts with canary testing
- Defined maintenance windows for updates



Physical Infrastructure

- Rigorous cable management and labeling
- Automated discovery and mapping of fabric
- Capacity planning with headroom for growth
- Failure scenario modeling and testing

Best Practices & Takeaways

1 Technology Selection

Choose Infiniband for stringent latency SLOs; RoCE for ecosystem flexibility and broader adoption—but engineer both with equal rigor

2 Co-designed Architecture

Align model parallelism strategies with underlying fabric topology and workload placement to minimize cross-node communication

3 Quality of Service

Enforce consistent QoS policies, align MTU/buffer configurations, and monitor **p99 latency metrics**—not just averages

4 Continuous Validation

Implement regular testing with both synthetic benchmarks (nccl-tests) and actual training workloads to catch performance regressions

